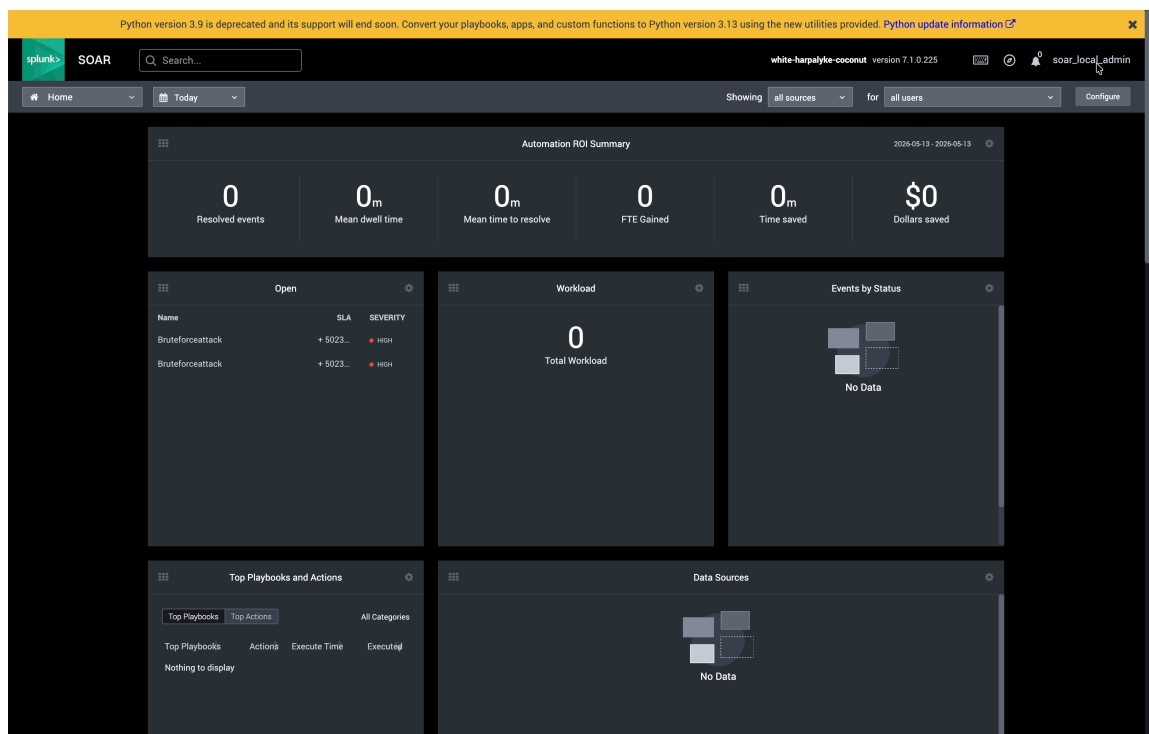


Creating a Privilege Escalation Playbook in SOAR

Summary

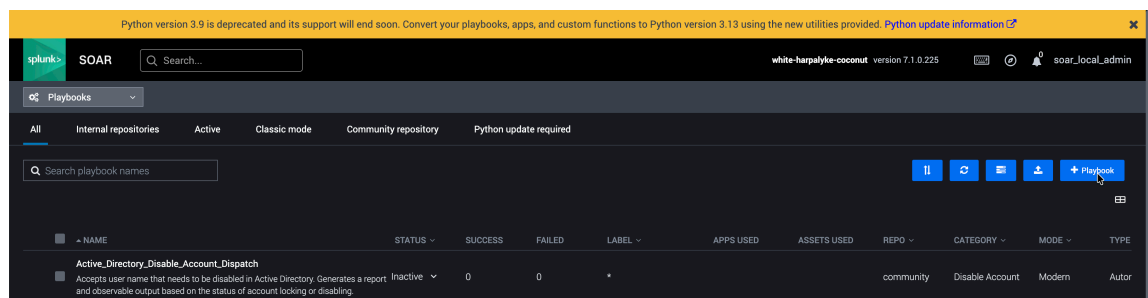
This document details the steps involved in Creating and Configuring the Privilege Escalation Use Case Playbook with all the required blocks added and configured and executed to validate the SOAR playbook by simulating the Privilege Escalation attack.

1. Login to Splunk SOAR as Admin



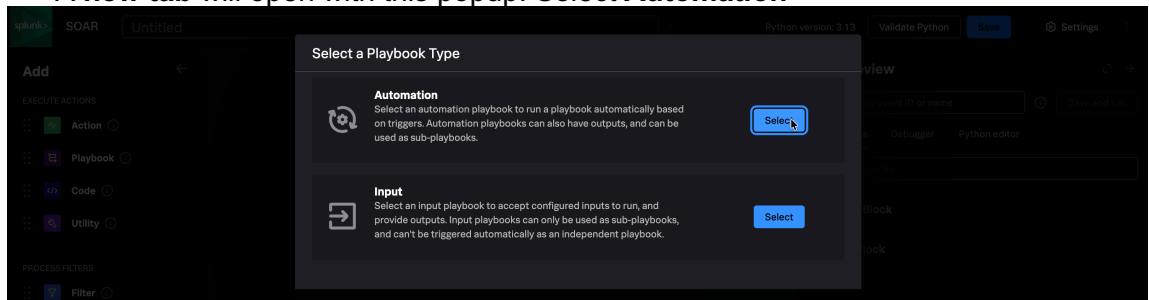
2. Create a new Playbook

1) Go to Home > Playbooks > + Playbook



2) Select Automation in Playbook Type popup

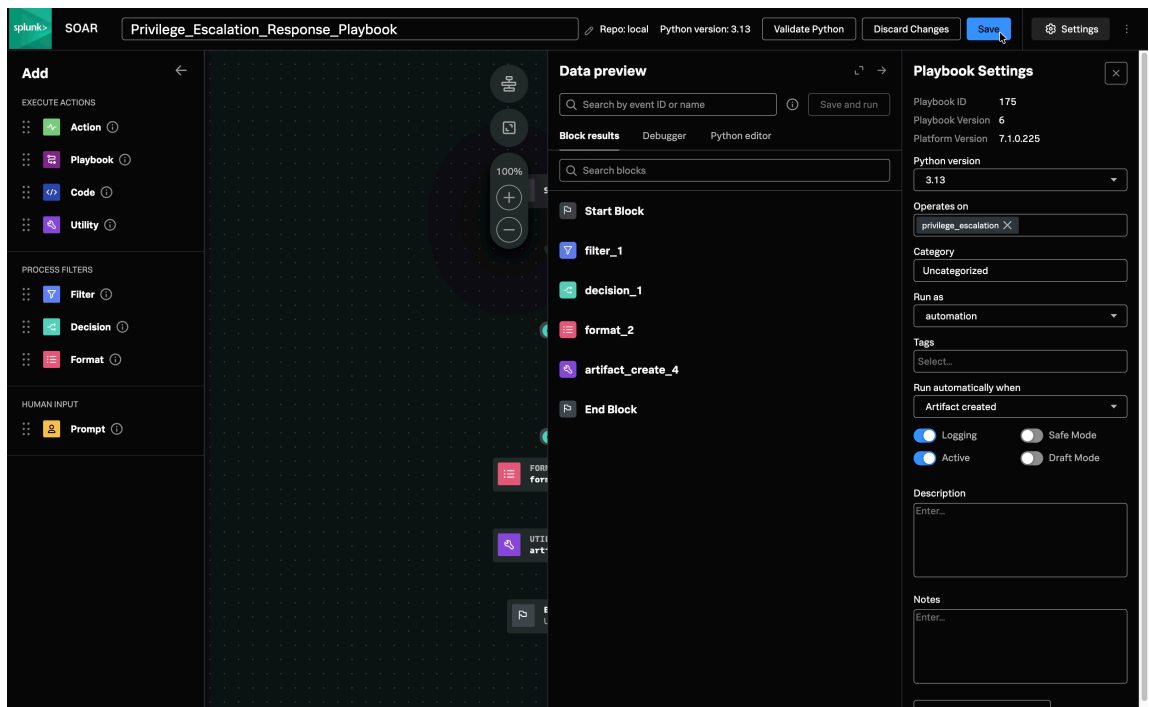
A new tab will open with this popup. Select **Automation**



3) Configure Playbook Settings

Click **Settings**,
Do the following Configuration and Save

Name: Privilege_Escalation_Response_Playbook
Operates on: privilege_escalation from drop-down
Run as: automation from drop-down
Run automatically when: Artifact created
Enable => Logging, Active



4) Save the Initial Config

Click **Save**

Repo to save to: local

Please enter a Comment (required): Initial Config of PrivEsc Playbook

Save Playbook

Repo to save to

local

Please enter a comment (required)

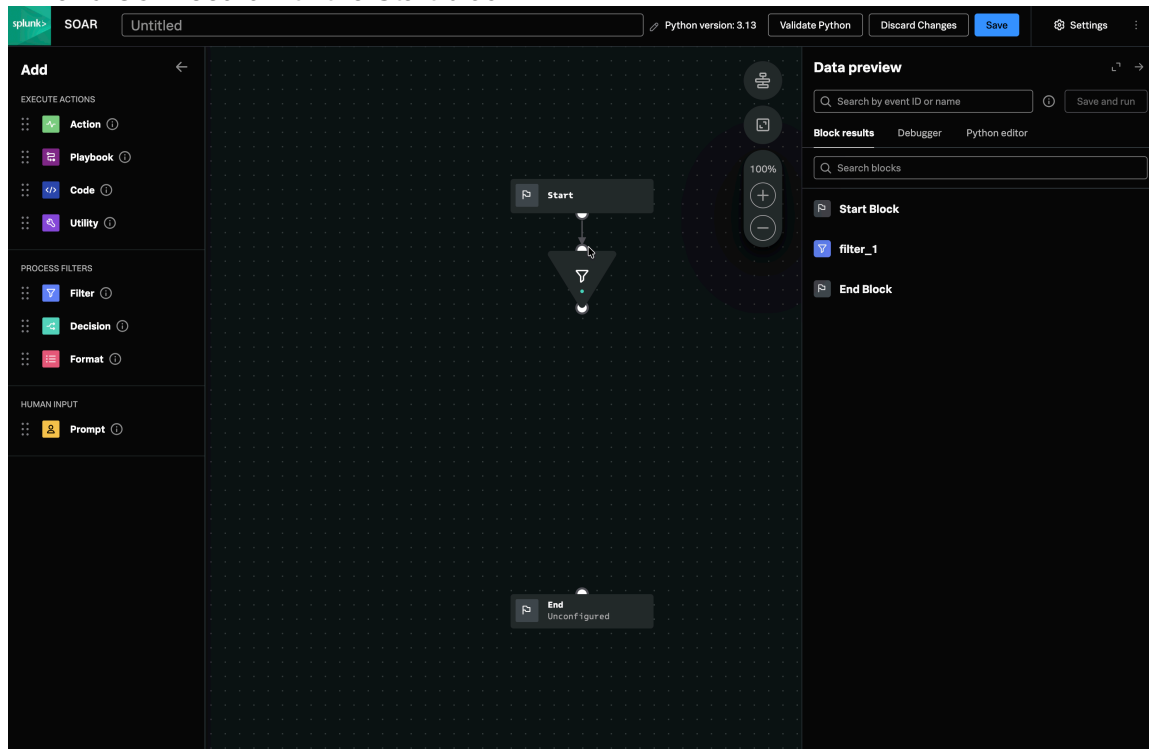
Initial Config of PrivEsc Playbook

☐ Reuse this comment until editor is reloaded (Ctrl + Save or Shift + Save button to show this dialog again)

Cancel
Save

3. Add Filter block and Configure

- 1) Click **x** to close the Playbook Settings panel on the right.
- 2) Drag and Drop **Filter block** on the playbook between **Start** and **End** blocks and **Connect** it with the Start block.



3) Configure the Filter block

- i) Enter the Condition to execute

Here **cs1** is the **group_name** value sent from the **Event Forwarding**.
This block filters out the events based on the **group_name** added.

Configure CONDITIONS

```
If
artifact*.cef.cs1 == Administrators
or
artifact*.cef.cs1 == Domain Admins
```

Click **Apply**

The screenshot displays the Splunk SOAR interface for configuring a Filter block. The 'Configure' panel on the left shows the 'CONDITIONS' section with an 'If' statement: 'artifact*.cefcs1 == Administrators' OR 'artifact*.cefcs1 == Domain Admins'. The 'Data preview' panel on the right shows the results of the filter, including a table with columns for 'filter_1: 1 item', 'comparisons: 2 items', and 'Results'.

- ii) **Save the changes to the Playbook**

Click **Save**

Repo to save to: local

Please enter a Comment (required): Added and Configured
Filter block in the playbook > Save

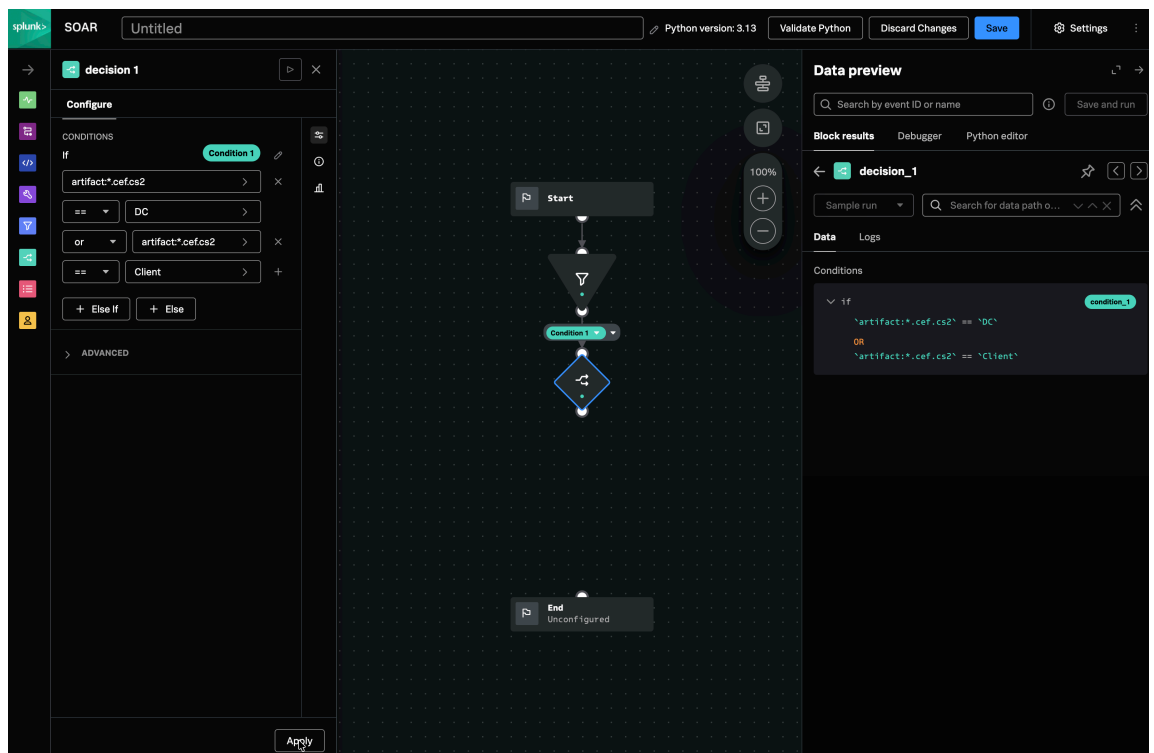
4. Add Decision block and Configure

- 1) Drag and drop the **Decision block** below the **Filter block** and **Connect** them.
- 2) **Configure the Decision block**
 - i) Enter the Condition to execute
Here **cs2** is the **source** value sent from the **Event Forwarding**.
This block decides if the event was performed on the source, **DC or Client**

Configure CONDITIONS

```
If
artifact*.cef.cs2 == DC
or
artifact*.cef.cs2 == Client
```

Click **Apply**



- ii) Save the changes to the Playbook
Click **Save**

Repo to save to: local

Please enter a Comment (required): Added and Configured Decision block in the PrivEsc Playbook. > Save

5. Add Format block and Configure

- 1) Drag and drop the **Format block** below the **Decision block** and **connect** them.
- 2) **Configure the Format block**
Format block organizes the data in a specific format and is configured as Input. It is later sent as output to the other blocks that are connected to it.

i) **Enter the following**

Configure:

```
<b>Potential privilege escalation
detected.</b><br><br>
<b>User SID:</b> {0}<br>
<b>Username:</b> {1}<br>
<b>Privileged Group:</b> {2}<br>
<b>Actor:</b> {3}<br>
<b>Host:</b> {4}<br>
<b>Source:</b> {5}
```

Parameters:

```
0: filtered-
data:filter_1:condition_1:artifact:*.cef.destinationU
serId
1: filtered-
data:filter_1:condition_1:artifact:*.cef.destinationU
serName
2: filtered-
data:filter_1:condition_1:artifact:*.cef.cs1
3: filtered-
data:filter_1:condition_1:artifact:*.cef.sourceUserNa
me
4: filtered-
data:filter_1:condition_1:artifact:*.cef.deviceHostna
me
5: filtered-
data:filter_1:condition_1:artifact:*.cef.cs2
```

Click **Apply**

The screenshot displays the SOAR interface for configuring a playbook named "Privilege_Escalation_Response_Playbook".

- Configuration Panel (Left):** Shows a template for a "Potential privilege escalation detected" event. It includes fields for User SID, Username, Privileged Group, and Actor, each with a corresponding "filtered-datafilter_1:col" dropdown menu.
- Flowchart (Center):** A visual representation of the playbook logic. It starts with a "Start" block, followed by a "Condition 1" block (which is currently set to "True"). This leads to a "Format 1" block, which then leads to an "End" block labeled "Unconfigured".
- Data Preview (Right):** Shows the output of the "format_1" block. It displays a list of filtered data items, each with a unique ID and a formatted string representing the event details.

ii) Save the changes to the Playbook

Click **Save**

Repo to save to: local

Please enter a Comment (required): Added and Configured Format block in the PrivEsc Playbook. > Save

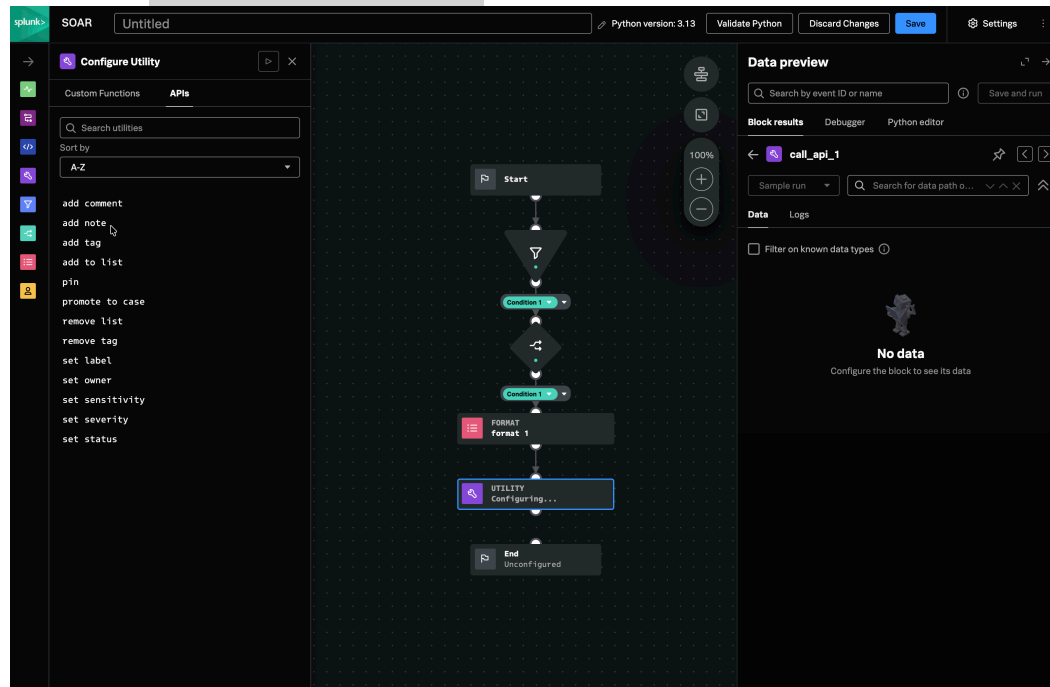
6. Add Utility blocks and Configure them

1) Add add note Utility block

i) Drag and drop the **Utility block** below the **Format block** and **connect** them.

ii) **Configure the Utility block**

Go to **APIs** tab > **add note**



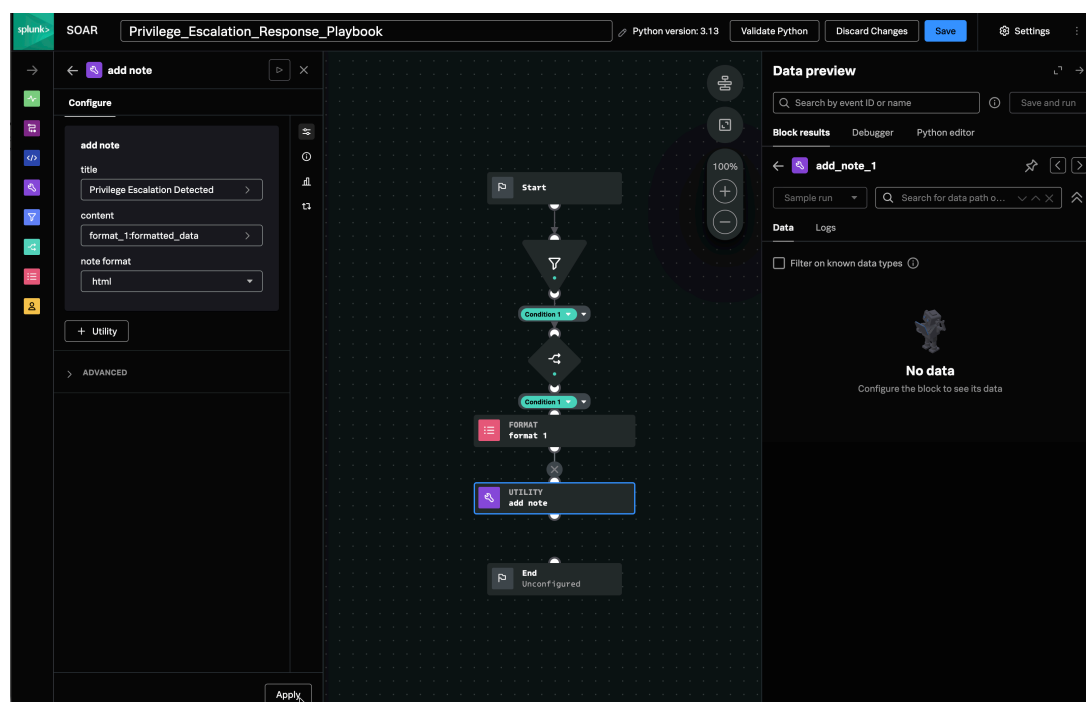
iii) Configure add note

title: Privilege Escalation Detected

content: format_1:formatted_data (select from the drop-down)

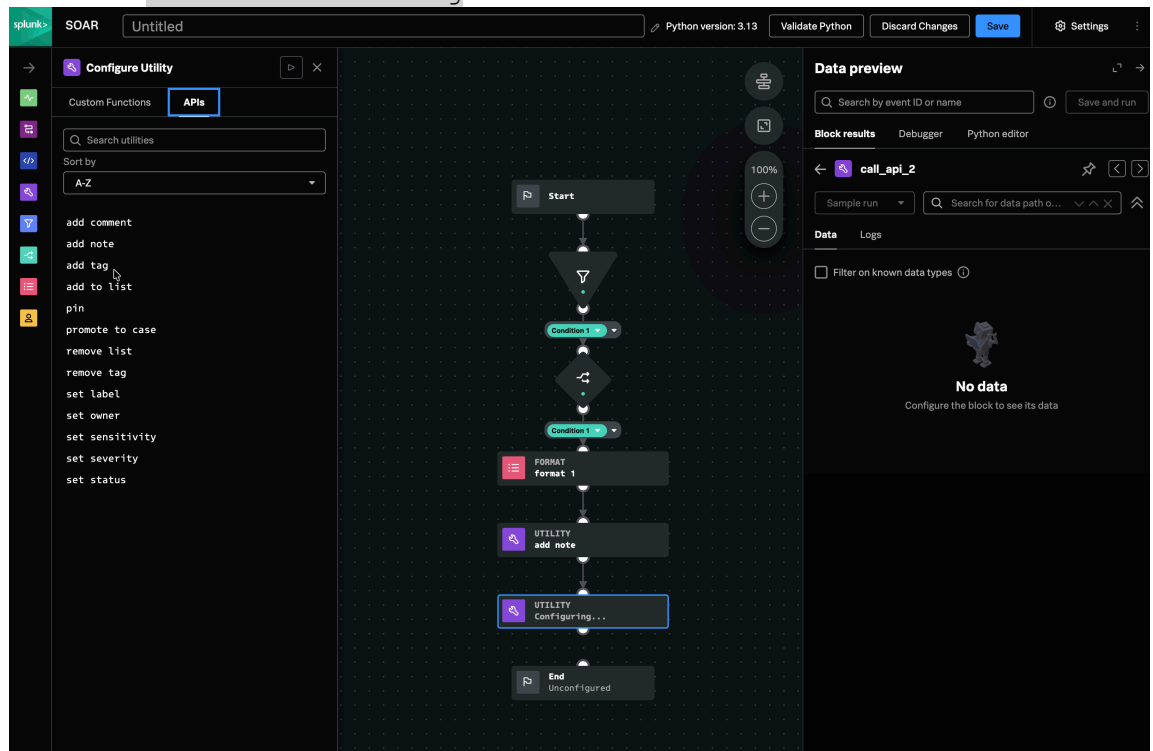
note format: html

Click **Apply**

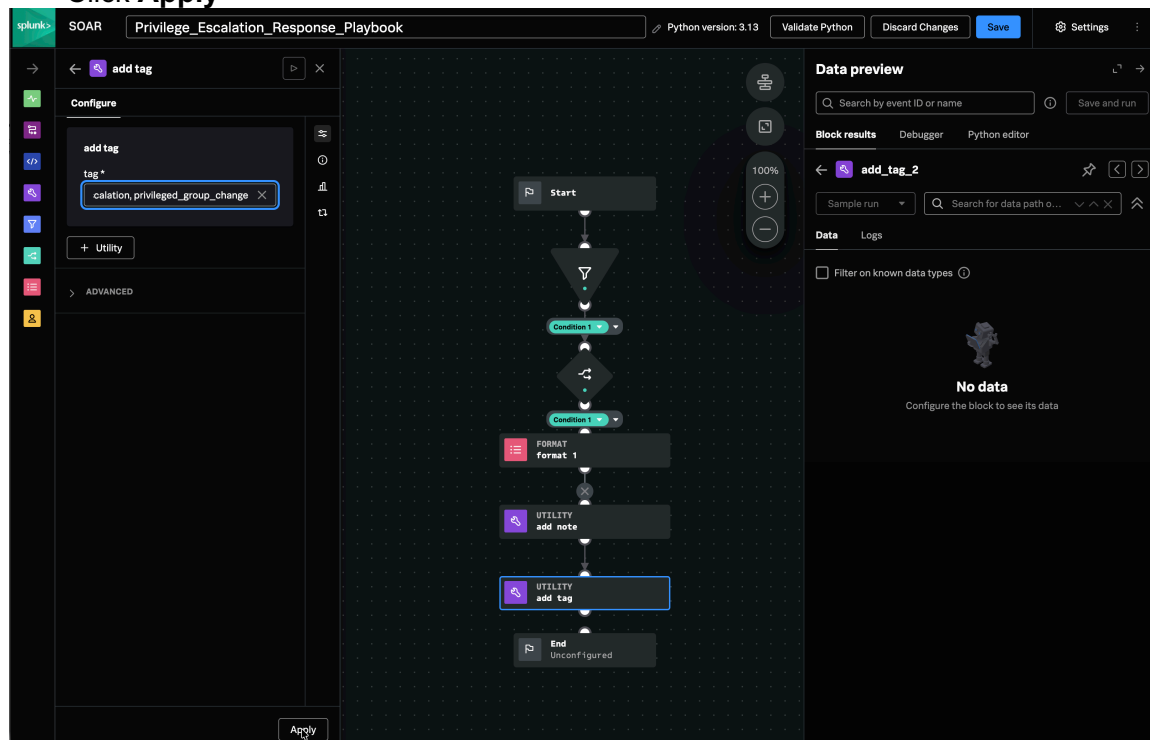


2) Add add tag Utility block

- i) Drag and drop **Utility block** below the **add note Utility block** and **connect** them.
- ii) **Configure the Utility block**
Go to **APIs** tab > **add tag**

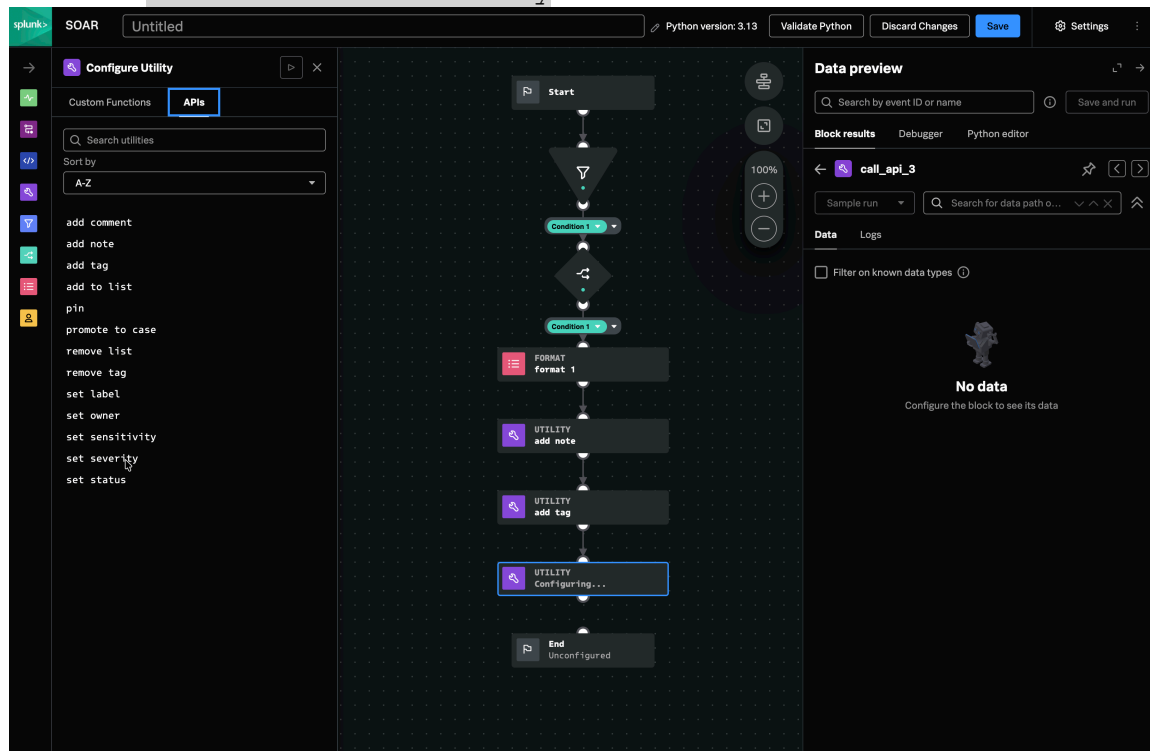


- iii) Configure add tag
tag: `privilege_escalation, privileged_group_change`
Click **Apply**



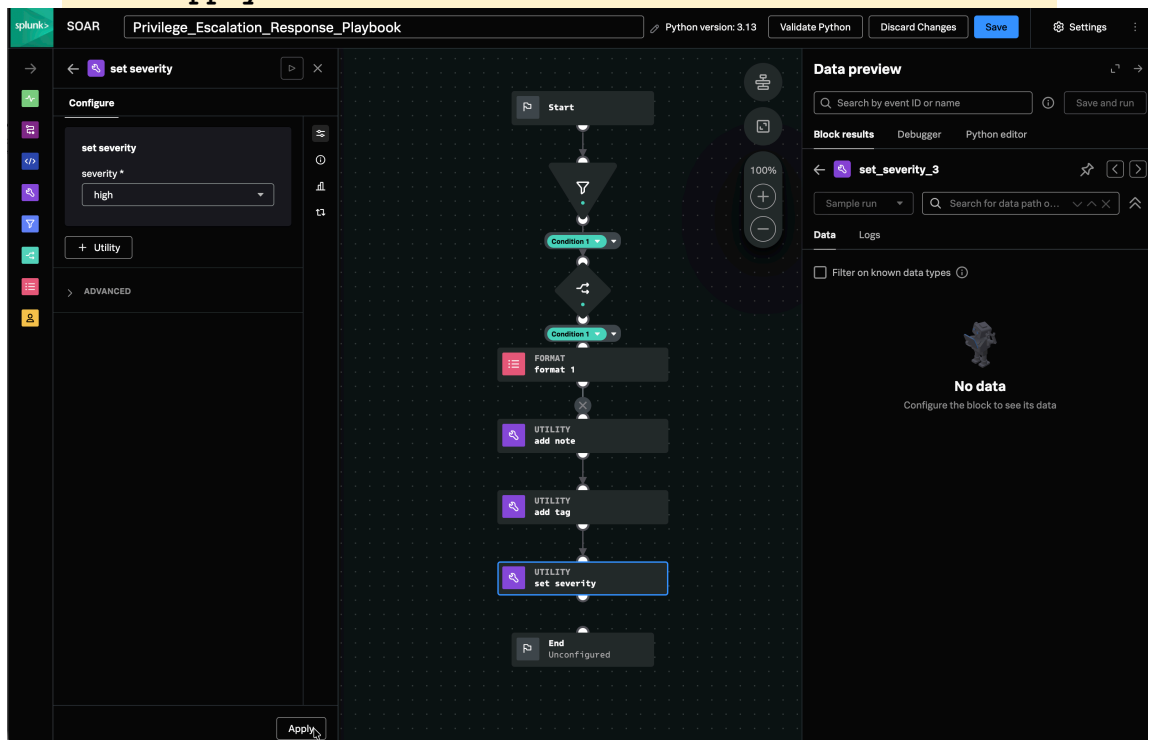
3) Add set severity Utility block

- Drag and drop **Utility block** below the **add tag Utility block** and **Connect** them.
- Configure the Utility block
Go to **APIs** tab > **set severity**



iii) Configure set severity

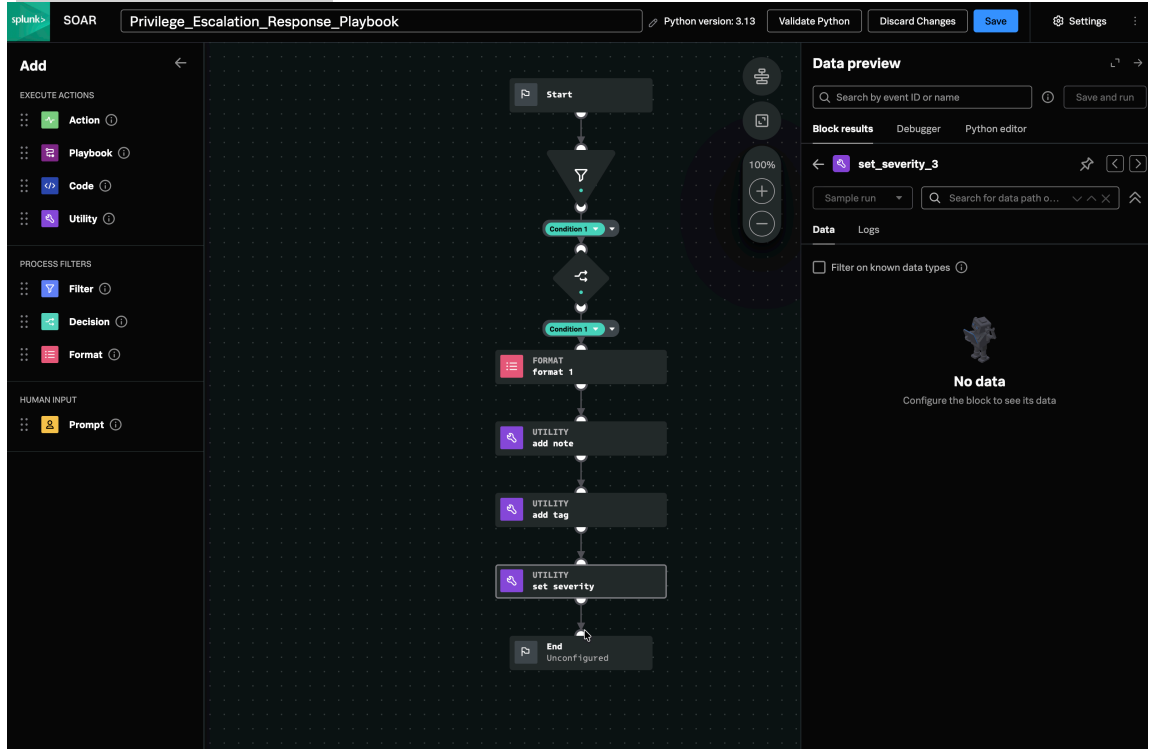
severity: high
Click **Apply** and **Connect** it to the **End block**.



7. Ensure Connections between the Blocks in the Playbook and Commit the changes

- 1) Ensure a proper **playbook title** is given. If not make Changes.
- 2) Ensure the **Playbook blocks are connected** in this way:

Start > Filter > Decision > Format > Add Note > Add Tag > Set Severity > End



- 3) **Save the Final Configuration Changes** to the Playbook

Save Playbook

Repo to save to

local

Please enter a comment (required)

Final Configuration of the PrivEsc Playbook

☐ Reuse this comment until editor is reloaded (Ctrl + Save or Shift + Save button to show this dialog again)

Cancel
Save

8. Test Playbook Execution

1) Ensure the Splunk Enterprise (Search Head) is connected to the SOAR Instance

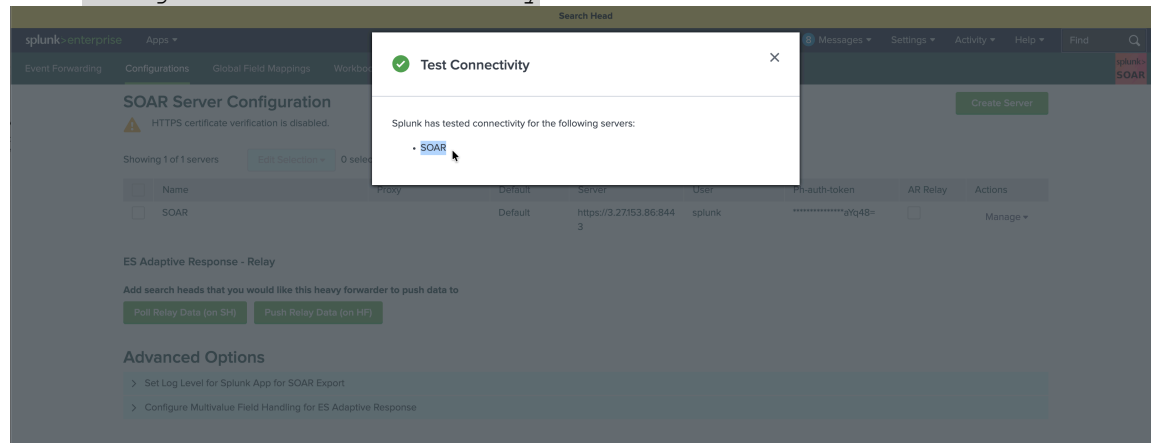
In **Search Head UI**,

Go to **Apps > Splunk App for SOAR Export app > Configurations** tab

Click on **Manage > Edit** and check the **Server** field.

Server: `https://<SOAR_PublicIP>:8443` > **Save**

Click **Manage > Test Connectivity**



2) Ensure the playbook is executed by doing a Preview in Search Head Event Forwarding

In **Event Forwarding** tab,

Click on the **Forwarder > Next > Save and Preview**

Specify a Time Range and Search

Click **Send to SOAR** on one of the Events

Preview for PrivEsc_Forwarding

Preview window

Last hour

Your search matched the following data

Successfully parsed CEF fields

Container Name

S-1-5-21-3021654831-3058487264-1625118876-1125

Sensitivity

amber

Severity

high

cs1

Administrators

cs2

Client

destinationUserId

S-1-5-21-3021654831-3058487264-1625118876-1125

deviceHostname

EC2AMAZ-CLIENT01.splunkadmin.local

externalId

4732

sourceUserName

admin

Send to SOAR

Your search matched the following data

Successfully parsed CEF fields

Container Name

S-1-5-21-3021654831-3058487264-1625118876-1125

Sensitivity

amber

Severity

high

cs1

Domain Admins

cs2

DC

destinationUserId

S-1-5-21-3021654831-3058487264-1625118876-1125

destinationUserName

eric wan

deviceHostname

EC2AMAZ-DC01.splunkadmin.local

externalId

4728

sourceUserName

admin

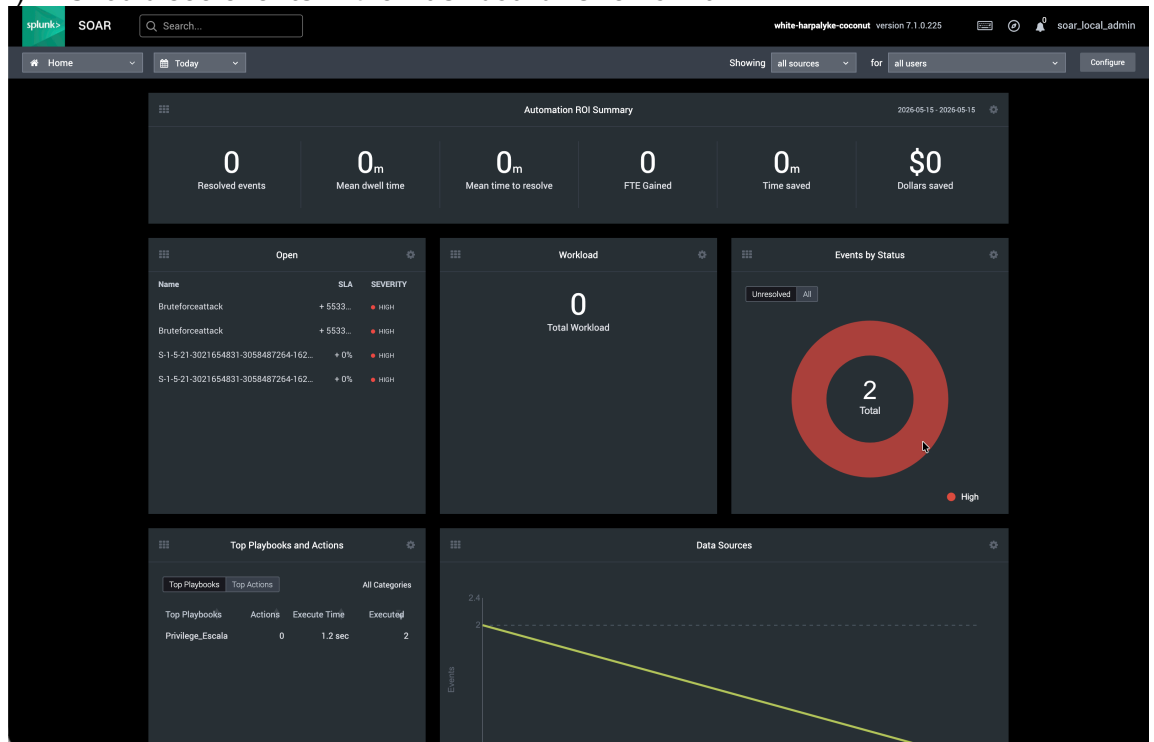
Send to SOAR

Done

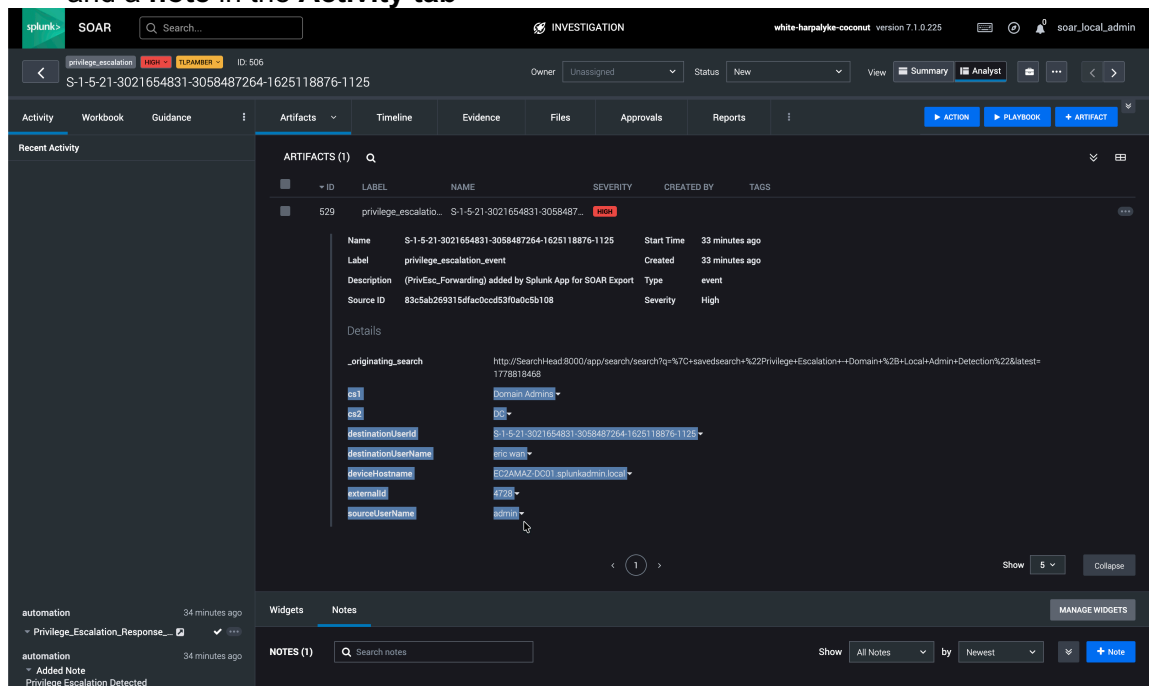
9. Ensure the Events are triggered by the Splunk alert via Event Forwarding in SOAR

In **SOAR UI**,

i) Should see **events** in the **Dashboard**. Click on it.



ii) In **Events page**, Click on a single event. Now You should see the **artifacts** and a **note** in the **Activity tab**



Notes tab

The screenshot shows the Splunk SOAR interface with the 'Notes' tab selected. The investigation ID is S-1-5-21-3021654831-3058487264-1625118876-1125. The 'Notes' tab displays a list of notes, with one note titled 'Privilege Escalation Detected' highlighted. The note details include:

- Title:** Privilege Escalation Detected
- General Note**
- by automation**
- May 15, 2026 4:15 am**
- Description:** Potential privilege escalation detected.
- User SID:** S-1-5-21-3021654831-3058487264-1625118876-1125
- Username:** eric wan
- Privileged Group:** Domain Admins
- Actor:** admin
- Host:** EC2AMAZ-DC01.splunkadmin.local
- Source:** DC

10. Simulate the PrivEsc attack and Test the Playbook Execution

- To see the entire workflow from Privilege Escalation Attack Simulation > Alert triggering > Event Forwarding > SOAR Playbook Execution, follow the steps referred in files - Refer files:
 - 23_Simulating_PrivEsc_Attack_WinServer_WinClient_V1.pdf and
 - 24_Logs_Validation_using_Extraction_Detection_for_ServerClient_Events_V1.pdf.
- For Validating SOAR Playbook Execution part, Refer Step 9 above.

Next Steps

For the Test Results and Observations – Refer 28_Test_Results_Observations_and_UseCase_Summary_PrivilegeEscalation_V1.pdf.